Domestic vs. Foreign Law: Portfolio Dynamics of Sovereign Debt

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  - Domestic law or foreign law.

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Our goal:

1. Develop a methodology to identify recovery rates from bond prices.
2. Explain why countries issue bonds under both laws.
3. Study how recovery rates affect welfare.

This paper

- **Empirics**
  - Document shares and duration under each law.
  - Compute yield gaps using local-currency bonds.
  - Estimate the sensitivity of yield gaps to credit risk.
- **Model**
  - Two bonds differ in duration and recovery rates.
  - Share and sensitivity identify the two recoveries.
- **Welfare**
  - Effects of raising the domestic-law recovery.

Application: Colombia

1. Recovery rates: 40% under domestic law, 47% under foreign law.
2. Trade-off: domestic-law debt is shorter, foreign-law debt recovers more.
3. $\uparrow$ Domestic-law recovery:
   - 12% of the gain from removing aggregate risk and default costs.

Related Literature

- **Quantitative sovereign default** (Aguiar & Gopinath 2006; Arellano 2008; Yue 2010; Arellano & Ramanarayanan 2012; Chatterjee & Eyigungor 2012; Hatchondo et al. 2016; Gordon & Guerrón-Quintana 2018; D'Erasmo & Mendoza 2021; Passadore & Xu 2022; Arellano, Mateos-Planas & Ríos-Rull 2023; Arellano & Barreto 2025; Passadore et al. 2026).
  - ↪ Two debts differ in recovery and maturity by **jurisdiction**.
- **Yield gaps** (Choi et al. 2011; Bradley et al. 2016; Du & Schreger 2016; Chamon et al. 2018; Randl & Zechner 2018; Schumacher et al. 2018; Saiegh & Biglaiser 2024).
  - ↪ Our approach allows us to analyze a new set of **emerging markets**.
- **Sovereign recovery rates** (Pan & Singleton 2008; Sturzenegger & Zettelmeyer 2008; Benjamin & Wright 2009; Cruces & Trebesch 2013; Asonuma & Trebesch 2016; Schlegl et al. 2019; Mihalache 2020; Dvorkin et al. 2021; Fang et al. 2021; Erce et al. 2022; Asonuma et al. 2023; Erce et al. 2024).
  - ↪ We estimate recovery rates for countries that have not **defaulted recently**.
- **Currency and investor composition** (Eichengreen & Hausmann 1999; Hausmann & Panizza 2003; Arslanalp & Tsuda 2014; Ottonello & Pérez 2019; Carstens & Shin 2019; Engel & Park 2022; Fang et al. 2025).
  - ↪ We emphasize the role of **governing law**, not currency or residency.

Empirics

Shares and duration: Colombian debt 2006–2024

	Median	Mean
Privately held debt / GDP (%)	20	21
Domestic law	**14**	15
Foreign law	**6**	6

+countries

	Min	Median	Mean
Duration at issuance (yrs)			
Domestic law	0.2	**3.8**	4.7
Foreign law	7.2	**8.7**	10.6

+countries

+data +shares +duration

Measuring the yield gap (Chamon et al. 2018)

1. Observe the **foreign-law** cashflows and yield ($y^F_{i,t}$).

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3. Construct a hypothetical yield ($\widetilde{y}^D_{i,t}$) from **cashflows** and **domestic-law** yield curve.

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2. Fit a **domestic-law** yield curve.
3. Construct a hypothetical yield ($\widetilde{y}^D_{i,t}$) from **cashflows** and **domestic-law** yield curve.
4. Compare the hypothetical yield with the observed yield:

$$\widetilde{y}^D_{i,t} - y^F_{i,t}$$

▸ more
▸ example
▸ equations

Yield Gap Rises with Default Risk

+countries

$$\Delta\left(\widetilde{y}^{D}_{i,t} - y^{F}_{i,t}\right) = \boldsymbol{\gamma}\,\Delta\mathrm{CDS}_t + \Gamma' X_{i,t} + \mu_i + u_{i,t}.$$

	Colombia
$\boldsymbol{\gamma}$	**0.227*****
	(0.050)
Law	New York
Observations	1,373
R^2	0.038
Window	2017–2020

Driscoll–Kraay SEs. $^{***}p<.01$.

A 100 bp increase in CDS spreads raises Colombia's yield gap by 23 bp.

Model

Recursive Formulation

Financial access:

$$\widehat{V}(y, b_D, b_F) = \max_{d \in \{0,1\}} \left\{ (1-d)\, V^R(y, b_D, b_F) + d\, V^D(y, b_D, b_F) \right\}$$

more

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Default:

$$V^D(y, b_D, b_F) = u(c) + \beta \mathbb{E}\Big[(1-\theta)\, V^D\big(y', b_D, b_F\big) + \theta\, \widehat{V}\big(y', \nu_D b_D, \nu_F b_F\big) \Big]$$

$$\text{s.t.} \quad c = \phi(y; d_0, d_1)$$

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$$\text{s.t.} \quad c = \phi(y; d_0, d_1)$$

Repayment:

$$V^R(y, b_D, b_F) = \max_{b'_D, b'_F} u(c) + \beta \mathbb{E}\left[\widehat{V}\left(y', b'_D, b'_F\right) \right]$$

$$\text{s.t.} \quad c + \lambda_D b_D + \lambda_F b_F = y + q_D^{ND}\left[b'_D - (1-\lambda_D) b_D \right] + q_F^{ND}\left[b'_F - (1-\lambda_F) b_F \right]$$

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Defaulted Debt Bond Prices, $i = D, F$

$$q_i^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\,\mathbb{E}\big[\, q_i^D\,(y', b_D, b_F)\big] + \theta\,\nu_i\, q_i^{ND}\,(y, \nu_D b_D, \nu_F b_F)$$

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Recovery rates (ν_i):

1. **Investors recover more.**
2. **Debt at re-entry is higher.**

New Issuance Bond Prices, $i = D, F$

$$q_i^{ND}\left(y, b_D', b_F'\right) = \frac{1}{1+r}\,\mathbb{E}\Big[\left(1-d\right)\chi + d\,q_i^{D}\left(y', b_D', b_F'\right)\Big]$$

$$\chi = \begin{cases} 1, & \lambda_i = 1, \\ \lambda_i + \left(1-\lambda_i\right) q_i^{ND}\left(y',\, b_D'', b_F''\right), & \lambda_i < 1. \end{cases}$$

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Long bonds ($\lambda_i < 1$):

1. **Insurance:** the market value of debt falls when output is low.

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Long bonds ($\lambda_i < 1$):

1. **Insurance:** the market value of debt falls when output is low.
2. **Dilution: future issuances** lower the price of longer debt today.

Calibration

more

Parameter	Target
Internally	
ν_D	**Foreign-law share**
ν_F	**Yield-gap sensitivity**
β	Debt / GDP
d_0	For-law spread, avg.
d_1	For-law spread, vol.
Externally	
λ_D	Duration: 3.8 yrs
λ_F	Duration: 8.7 yrs

Calibration

more

Parameter	Value
Internally	
ν_D	0.40
ν_F	0.47
β	0.92
d_0	−0.19
d_1	0.25
Externally	
λ_D	0.25
λ_F	0.10

Moment	Data	Model
Targeted		
Foreign-law share	**29%**	**29%**
Yield-gap sensitivity	**+0.23**	**+0.23**
Debt / GDP	20.9%	21.0%
For-law spread, avg.	2.40%	2.34%
For-law spread, vol.	0.75%	1.02%
Untargeted		
corr(For-law issuance/issuance, y)	−0.29	−0.69
$\sigma(c)/\sigma(y)$	1.15	1.17
$\mathrm{corr}(c, y)$	0.92	0.83
Avg. debt service / GDP	3.9%	4.3%
Default probability	—	3.7%

Dilution-default trade-off

Recovery differences:

1. Investors recover more.
2. Re-entry debt is higher.

Dilution-default trade-off

costs

Recovery differences:

1. Investors recover more.
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Experiment:

- Same durations
- Different recoveries

Result:

- Use the **high-recovery** asset

more

$\Rightarrow$ **Foreign-law bonds** protect against default risk.

Dilution-default trade-off

costs

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Result:

- Use the **high-recovery** asset

more

$\Rightarrow$ **Foreign-law bonds** protect against default risk.

Duration differences:

1. Insurance.
2. Dilution.

Dilution-default trade-off

costs

Recovery differences:

1. Investors recover more.
2. ~~Re-entry debt is higher.~~

Experiment:

- Same durations
- Different recoveries

Result:

- Use the **high-recovery** asset

more

⇒ **Foreign-law bonds** protect against default risk.

Duration differences:

1. ~~Insurance.~~
2. Dilution.

Experiment:

- Same recoveries
- Different durations

Result:

- Use the **shorter** asset

more

⇒ **Domestic-law bonds** protect against dilution risk.

Welfare

Domestic-Law Recovery (0.40 $\rightarrow$ 0.47)

- Removing aggregate risk and default costs: **+71 bp**.

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	Δ spread (bp)		Δ Debt/GDP	Δ Def. prob.
	Domestic	Foreign	(bp)	(bp)
$\uparrow \nu_D$	−46.3	−22.4	+40	−30

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$\uparrow \nu_D$	−46.3	−22.4	+40	−30	**+8.2**

- **Welfare gains:**
  - **∼12%** of removing aggregate risk and default costs.

more

Domestic-Law Recovery (0.40 → 0.47)

- Removing aggregate risk and default costs: **+71 bp**.

	Δ spread (bp)		Δ Debt/GDP	Δ Def. prob.	**Δ welfare**	of which (bp)		
	Domestic	Foreign	(bp)	(bp)	**(bp)**	Prices	Re-entry	Behaviour
$\uparrow \nu_D$	−46.3	−22.4	+40	−30	**+8.2**	+2.1	−1.3	+7.4

- **Welfare gains:** more
  - **∼12%** of removing aggregate risk and default costs.
  - Mostly driven by changes in behaviour, not prices. more
- **Robustness:** patient households and different levels of initial debt. more

Thank you!

Appendix

Data sources

back

- **Dealogic:** ISINs, maturity, coupon dates, foreign law.
- **Prospectuses:** amount issued, bond terms, foreign law.
- **J.P. Morgan EMBI Global:** sovereign spreads, foreign law.
- **Central bank / MoF:** yield curve, auctions, bond terms, domestic law.
- **Bloomberg:** bond prices, bid–ask spreads, both laws.
- **Markit:** 5-year sovereign CDS spreads.
- **World Bank / FRED / IMF:** macro aggregates, exchange rates, risk-free rates.

Shares: Privately held debt / GDP

back

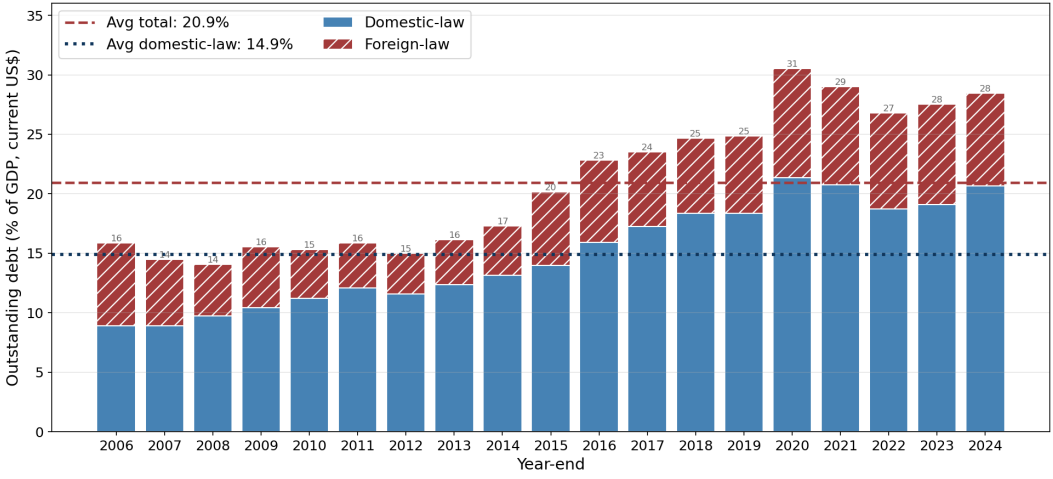
- **Scope:** marketable central-government securities priced by private investors, i.e. auction-placed TES (*Subastas administradas*) + foreign-law global bonds. Excludes non-auction placements to public entities (*convenidas*), official loans, and central-bank holdings.
- **Stock:** year-end outstanding stock, by governing law.
- **Debt / GDP** (current-FX + UVR valuation):

$$\left(\tfrac{D}{Y}\right)_t = \frac{D_t^{\text{dom}}(\text{TRM}_t^{\text{avg}}, \text{UVR}_t^{\text{avg}}) + D_t^{\text{fgn},\$}}{\text{GDP}_t^{\$}}, \qquad \text{GDP}^{\$} = \texttt{NY.GDP.MKTP.CD}.$$

  (fixed-rate: COP nominal; UVR-indexed: UVR nominal $\times$ annual-average UVR index) $\rightarrow$ USD at the annual-average TRM. Foreign tranches are already in USD.
- **Foreign-law share:** $\text{share}_t^{\text{fgn}} = D_t^{\text{fgn}} / \left(D_t^{\text{fgn}} + D_t^{\text{dom}}\right)$, same stocks and valuation. Falls from $\approx 44\%$ (2006) to $\approx 27\%$ (2024), averaging $\approx 29\%$.

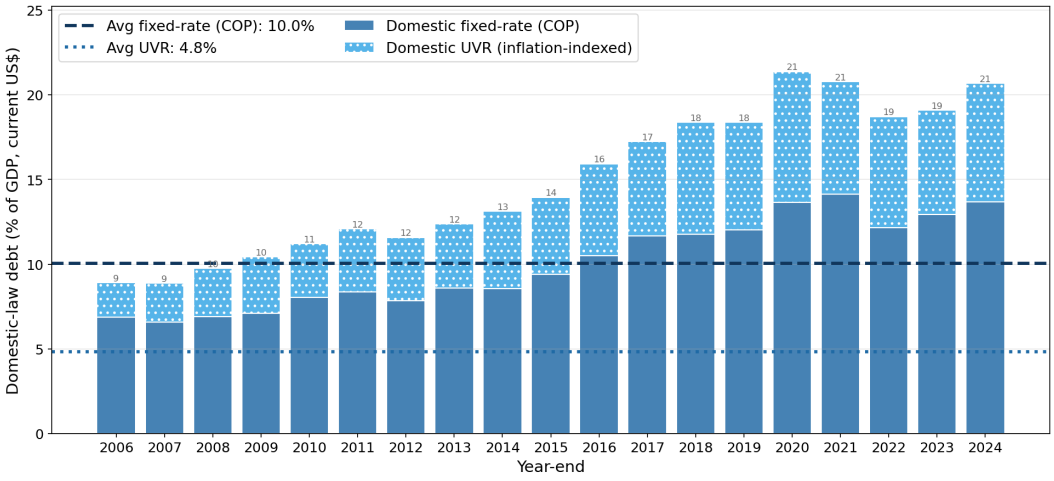
Shares: over time

back



Shares: Domestic-Law Debt, Inflation-Indexed (UVR) vs. Fixed-Rate

back



Shares: Foreign participation in domestic-law debt

back

Denominator	Latest	Mean	Median
Total general *(reported)*	19.6%	17.3%	20.7%
Total – central bank (BanRep)	20.8%	18.2%	21.8%
Total – core public sector	21.9%	20.0%	23.3%
Total – core public – Fiducia Pública	23.2%	22.9%	26.5%

Shares: Foreign-Law Share Across Countries

back

	Mean	Median
Brazil	5.0	4.7
Chile	23.8	19.2
Colombia	**29.7**	**30.1**
Hungary	30.5	31.6
Mexico	16.0	16.0
Paraguay	51.9	72.6
Peru	34.7	35.4
Poland	20.8	20.5
Russia	21.1	21.5
South Africa	8.5	8.3
Turkey	24.5	22.5
Uruguay	62.5	63.5

Share of privately held debt issued under foreign law, 2006–2024 year-end stock.

Duration: Duration by Law Across Countries

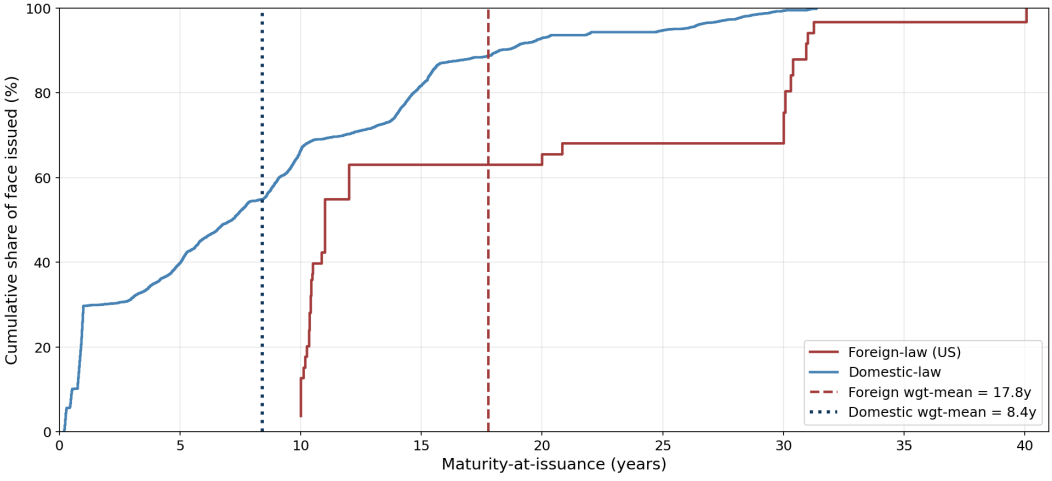
back

	Median		Mean	
	Foreign	Domestic	Foreign	Domestic
Brazil	8.40	3.47	9.65	4.35
Chile	9.90	8.85	12.13	10.31
Colombia	**8.71**	**3.75**	**10.56**	**4.74**
Hungary	7.54	2.84	7.60	3.74
Mexico	8.87	0.48	10.14	2.67
Paraguay	8.74	3.46	10.54	3.88
Peru	11.77	8.64	13.46	9.28
Poland	8.19	2.56	8.75	3.97
Russia	8.00	5.80	8.98	5.49
South Africa	8.79	6.88	10.12	6.89
Turkey	7.64	1.43	7.83	2.25
Uruguay	9.45	8.19	10.55	8.55

Macaulay duration at issuance, in years.

Duration: Cumulative face issued by maturity at issuance

back



Duration: Maturity buckets, share of face value issued

back

Maturity (yr)	Foreign $bn	Foreign %	Domestic $bn	Domestic %
0–1	0.0	0.0	64.0	29.7
1–5	0.0	0.0	21.6	10.0
5–10	1.5	3.8	57.2	26.5
10–20	23.6	59.3	57.5	26.7
20–30	3.6	9.2	13.9	6.5
30+	11.0	27.8	1.3	0.6
Total	39.8	100.0	215.5	100.0

Duration: Maturity and duration buckets, GDP-weighted shares

back

	Maturity		Duration	
Bucket	Domestic	Foreign	Domestic	Foreign
$\leq$ 1y	32.0	0.0	31.7	0.0
1–3y	2.1	0.0	4.2	0.0
3–5y	8.7	0.0	10.2	0.0
5–10y	25.3	4.1	40.3	65.1
$>$ 10y	31.9	95.9	13.5	34.9
Median (yrs)	4.08	11.00	**3.75**	**8.71**
Mean (yrs)	6.64	17.79	4.74	10.56

Duration: Maturity and duration by governing law

back

	Maturity (yrs)		Duration (yrs)	
	Foreign	Domestic	Foreign	Domestic
Min	10.00	0.21	7.23	0.21
p10	10.10	0.31	7.64	0.31
p25	10.37	0.83	8.20	0.84
Median	11.00	4.08	8.71	3.75
p75	30.00	10.12	12.50	7.94
p90	30.95	16.63	15.72	10.78
Max	40.07	31.37	20.49	18.95
Mean	17.79	6.64	10.56	4.74
Face-weighted mean	17.76	8.41	10.58	5.79

Duration: Issuance size by governing law

back

% of current GDP

	Foreign-law (US)	Domestic-law
min	0.1481	0.0003
p10	0.3068	0.0093
p25	0.3699	0.0122
median	0.4644	0.0188
p75	0.5315	0.0274
p90	0.6041	0.0419
max	0.6279	0.1831
mean	0.4506	0.0227

USD amounts

	Foreign-law (US)	Domestic-law
n issuances	29	3,125
total (USD bn)	39.76	215.47
min (USD mn)	548.85	0.81
p10 (USD mn)	1,000.0	24.7
p25 (USD mn)	1,000.0	32.4
median (USD mn)	1,500.0	52.5
p75 (USD mn)	1,600.0	86.9
p90 (USD mn)	2,000.0	131.4
max (USD mn)	2,200.0	699.7
mean (USD mn)	1,371.0	69.0

Duration: Computing Macaulay duration at issuance

back

Observe (from the bond's terms):

- Coupon c, frequency m (payments/yr), years to maturity N.
- Own yield at issuance y (effective-annual):
  - **Foreign-law (US):** reoffer YTM; coupon/frequency from the tranche terms.
  - **Domestic-law:** price-implied YTM, from the issue price and (annual) coupon.

Compute (build the cash-flow schedule, then discount):

- Dates: $t_i = i/m, \;\; i = 1, \ldots, n, \;\; n = \mathrm{round}(N\,m)$.
- Cash flows: $c_i = c/m$; add principal $(= 1)$ at maturity, $c_n = c/m + 1$.

$$D \;=\; \frac{\sum_{i=1}^{n} t_i \, c_i/(1+y)^{t_i}}{\sum_{i=1}^{n} c_i/(1+y)^{t_i}}.$$

Get (conventions & checks):

- Zero-coupon instruments and bills: $D = N$.
- Floaters/zeros (no fixed coupon or reoffer YTM): in maturity, not *duration*.
- Validation: price-implied YTM $\approx$ auction cutoff.

The Yield Gap in Every Country

back

$$\Delta\left(\tilde{y}^{D}_{i,t} - y^{F}_{i,t}\right) = \boldsymbol{\gamma}\,\Delta\mathrm{CDS}_t + \Gamma' X_{i,t} + \mu_i + u_{i,t}.$$

	Colombia	Brazil	Paraguay	Russia	Peru *(placebo)*
γ	0.227***	0.664***	0.157**	0.076***	0.135
	(0.050)	(0.091)	(0.072)	(0.023)	(0.087)
Law	US	US	US	UK	Peru
Observations	1,373	3,399	713	653	652
R^2	0.038	0.096	0.042	0.072	—
Window	2017–2020	2014–2020	2025–2026	2014–2017	2017–2020

First differences; Driscoll–Kraay SEs in parentheses, HC1 for Peru. $^{*}p<.10$, $^{**}p<.05$, $^{***}p<.01$.

Calibration: All Parameters

moments back

Parameters

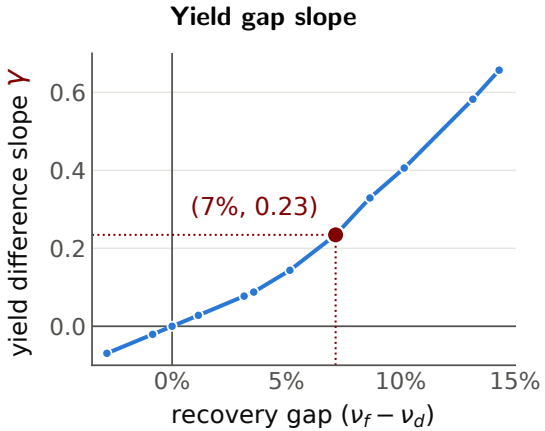
Parameter	Value	Target
Externally calibrated		
γ	2.00	Standard
θ	0.25	Four-year exclusion
r	0.02	US risk-free rate
ρ	0.77	Output process
σ_ε	0.03	Output process
λ_D	0.25	Duration: 3.8 yrs
λ_F	0.10	Duration: 8.7 yrs
Internally calibrated		
β	0.92	Debt / GDP
d_0	-0.19	For-law spread, avg.
d_1	0.25	For-law spread, vol.
ν_D	0.40	**Foreign-law share**
ν_F	0.47	**Yield-gap sensitivity**

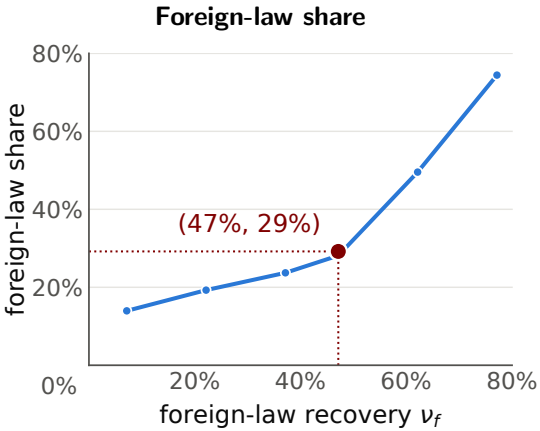
Moments

Moment	Data	Model
Targeted		
Foreign-law share	**29%**	**29%**
Yield-gap sensitivity	**+0.23**	**+0.23**
Debt / GDP	20.9%	21.0%
For-law spread, avg.	2.40%	2.34%
For-law spread, vol.	0.75%	1.02%
Untargeted		
corr(For-law issuance/issuance, y)	-0.29	-0.69
$\sigma(c)/\sigma(y)$	1.15	1.17
$\mathrm{corr}(c, y)$	0.92	0.83
Avg. debt service / GDP	3.9%	4.3%
Default probability	—	3.7%

How the Two New Moments Move

back





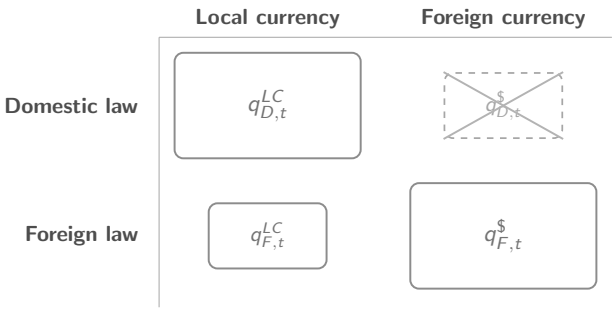
Measuring the yield gap between laws

back

	Local currency	Foreign currency
Domestic law	$q^{LC}_{D,t}$	$q^{\$}_{D,t}$
Foreign law	$q^{LC}_{F,t}$	$q^{\$}_{F,t}$

Measuring the yield gap between laws

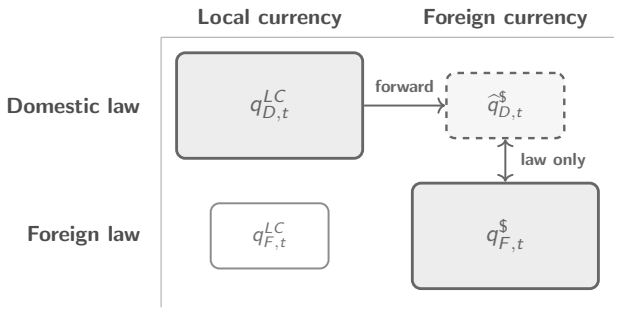
back



- **Challenge:**
  - Most EMEs do not issue domestic-law debt in foreign currency.
  - Relatively few bonds in local currency under foreign law.

Measuring the yield gap between laws

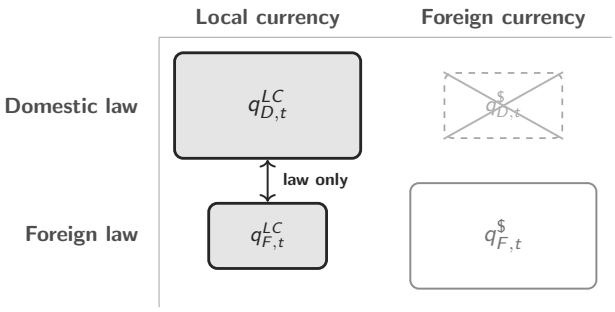
back



- **Existing evidence:** Chamon et al. (2018)
  - Build the missing bond using currency forwards.
  - Requires assuming no depreciation upon default.
  - Restricts the approach to euro-area countries.

Measuring the yield gap between laws

back



- **This paper:**
  - Fit the domestic-law yield curve using all domestic-law bonds.
  - Reprice the foreign-law bond's outstanding cashflows on that curve.
  - Compare the foreign-law yield to the yield that bond would carry under domestic law.

An example: Colombia's 9.85% bond

back

- At $t = 11/02/18$ it trades at $q^{LC}_{F,t} = 126.68$, with a yield of $y_{F,t} = 6.28\%$.

	06/28/19	06/28/20	$\cdots$	06/28/27
Cash flow	9.85	9.85	$\cdots$	109.85
Time until payment (yrs)	0.7	1.7	$\cdots$	8.7
Domestic-law yield at t (%)	4.62	5.19	$\cdots$	7.00
Value at t	9.57	9.06	$\cdots$	61.18

Domestic-law price:

$$\widetilde{q}^{LC}_{D,t} = 9.57 + 9.06 + \cdots + 61.18 = 123.01 \implies \widetilde{y}_{D,t} = 6.77\%$$

Yield gap$_t$:

$$\widetilde{y}_{D,t} - y_{F,t} = 6.77\% - 6.28\% \approx \textbf{50 bp}$$

equations

Fitting the domestic-law yield curve

back

Model: Nelson–Siegel zero-coupon curve, re-estimated every day

$$y_t(\tau) = \beta_{0,t} + \beta_{1,t}\,\frac{1 - e^{-\tau/\lambda_t}}{\tau/\lambda_t} + \beta_{2,t}\left(\frac{1 - e^{-\tau/\lambda_t}}{\tau/\lambda_t} - e^{-\tau/\lambda_t}\right)$$

- β_0 is the level, β_1 the slope and β_2 the curvature.
- λ sets where the hump sits, measured in years.
- $y_t(\tau)$ is the annually compounded zero rate at maturity τ.

Estimation

- Price each bond off the curve: $\hat{P}_{j,t} = \sum_k CF_{j,k}\,\big(1 + y_t(\tau_k)\big)^{-\tau_k}$.
- Choose β to minimise price errors in yield units: $\min_\beta \sum_j \big[(\hat{P}_{j,t} - P_{j,t})/\mathrm{DV01}_{j,t}\big]^2$.

How λ is chosen

back

Pricing error of bond j on day t:

$$e_{j,t}(\beta,\lambda) = \frac{\hat{P}_{j,t}(\beta,\lambda) - P_{j,t}}{\mathrm{DV01}_{j,t}}, \qquad \mathrm{SSE}_t(\beta,\lambda) = \sum_j e_{j,t}(\beta,\lambda)^2.$$

Step 1: best λ each day. For each λ on a grid of 16 values from 0.15 to 3 years, solve $\min_\beta \mathrm{SSE}_t(\beta,\lambda)$; keep the λ with the lowest minimised SSE_t. This gives $\lambda_t^{\mathsf{free}}$.

Step 2: smooth λ, then re-estimate β.

$$\lambda_t^{\mathsf{smooth}} = \mathrm{median}\{\lambda_{t-10}^{\mathsf{free}}, \ldots, \lambda_{t+10}^{\mathsf{free}}\}, \qquad \hat{\beta}_t^s = \arg\min_\beta \mathrm{SSE}_t(\beta, \lambda_t^s), \quad s \in \{\mathsf{free}, \mathsf{smooth}, \mathsf{fixed}\}.$$

Check: does smoothing cost fit? Compare three λ series, free, smooth, and fixed, each at its $\hat{\beta}_t$:

- **Fit:** median over days of $\sqrt{\mathrm{SSE}_t/N_t}$, the typical pricing error per bond (bp).
- **Stability:** per tenor, median of $|y_t(\tau) - y_{t-1}(\tau)|$ across dates; then average across tenors.
- **Rule:** among the series whose fit is within 25% of the best, pick the most stable.

How to compare bonds with different maturities and coupons?

back

- Foreign-law bonds have a daily price $q^{LC}_{F,t}$, payments $c_1, \ldots, c_n$ on dates $s_1 < \cdots < s_n$.
- At date t, the domestic-law yield at tenor $(s_k - t)$ is $z_t(s_k - t)$.

1. Given $q^{LC}_{F,t}$ and the remaining sequence of coupons $(c_k)_{\{k:\, s_k \geq t\}}$, solve for $\boldsymbol{y_{F,t}}$:
$$q^{LC}_{F,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + \boldsymbol{y_{F,t}}\right)^{s_k - t}}.$$
2. Given domestic-law yields and the remaining coupons, solve for $\boldsymbol{\widetilde{q}^{LC}_{D,t}}$:
$$\boldsymbol{\widetilde{q}^{LC}_{D,t}} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + z_t(s_k - t)\right)^{s_k - t}}.$$
3. Given that price $\widetilde{q}^{LC}_{D,t}$ and the remaining coupons, solve for $\boldsymbol{\widetilde{y}_{D,t}}$:
$$\widetilde{q}^{LC}_{D,t} = \sum_{\{k:\, s_k \geq t\}} \frac{c_k}{\left(1 + \boldsymbol{\widetilde{y}_{D,t}}\right)^{s_k - t}}.$$
4. **Yield gap**$_t = \boldsymbol{\widetilde{y}_{D,t}} - \boldsymbol{y_{F,t}}$: two identical bonds except for the law.

Selective default across laws is rare

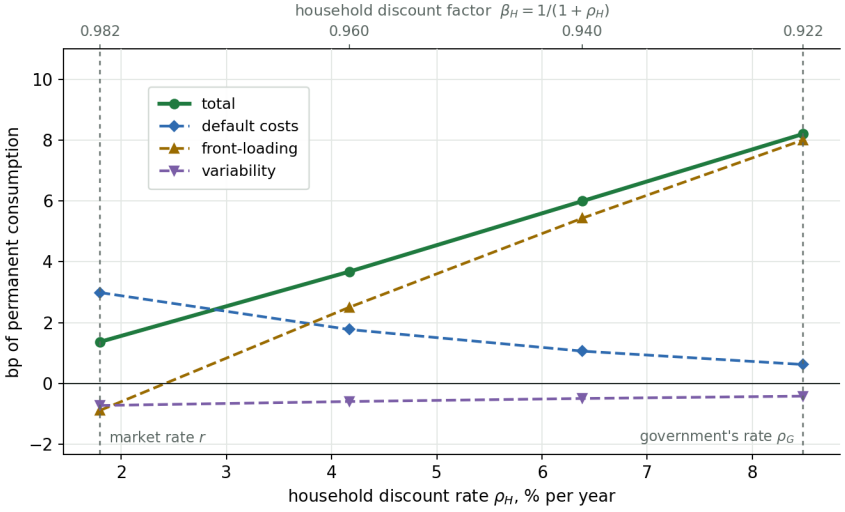
back

Sovereign bond defaults on private creditors, 2000–2018 (Erce, Mallucci & Picarelli)

Bond restructurings	32
with bonds under both laws	19
Of those 19:	
Not selective	8
One class spared, but not by law	8
spared bonds held by domestic banks	3
spared bonds lost more to inflation	2
spared bonds held by the state	1
spared bonds held by captive pension funds	1
carve-out by residency, not law	1
Selective	**1**

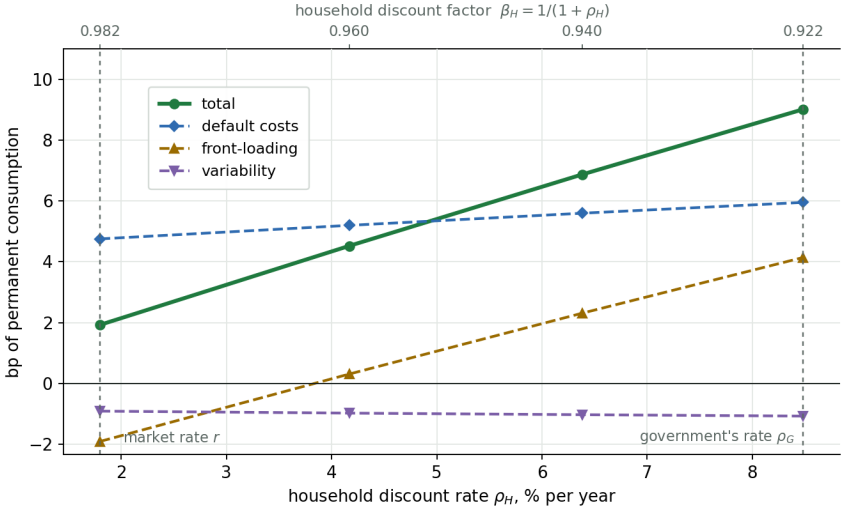
Welfare with More Patient Households

back



Welfare at Mean Debt

back



How the three effects are computed

back

$\bar{V}^E$ the equilibrium value: c with market access, $\phi(y)$ while excluded.

$\bar{V}^{ND,E}$ the same recursion with $c = y$ instead of $\phi(y)$ while excluded: write-down $\nu_i b_i$ kept.

$\bar{V}(\bar{c}^{ND,E})$ the mean path of c^{ND} for sure, $\sum_t \beta_H^t u(\bar{c}_t^{ND,E})$: output costs removed.

$\bar{V}^A$, $\bar{V}^A(y^\infty)$ autarky forever at the risky endowment, and at the constant mean $y^\infty = \sum_y \pi(y)\, y$.

Value of market access, and its exact split (Aguiar–Amador–Fourakis, 2020)

$$1 + \chi^E = \left(\frac{\bar{V}^E}{\bar{V}^A}\right)^{\frac{1}{1-\gamma}} = (1 + \chi_D^E)(1 + \chi_V^E)(1 + \chi_T^E),$$

$$\underbrace{1 + \chi_D = \left(\frac{\bar{V}}{\bar{V}^{ND}}\right)^{\frac{1}{1-\gamma}}}_{\text{default costs}} \qquad \underbrace{1 + \chi_V = \left(\frac{\bar{V}^{ND}}{\bar{V}(\bar{c}^{ND})} \cdot \frac{\bar{V}^A(y^\infty)}{\bar{V}^A}\right)^{\frac{1}{1-\gamma}}}_{\text{variability}} \qquad \underbrace{1 + \chi_T = \left(\frac{\bar{V}(\bar{c}^{ND})}{\bar{V}^A(y^\infty)}\right)^{\frac{1}{1-\gamma}}}_{\text{front-loading}}$$

What the figure plots. Reform E_1 ($\nu_D = 0.4715$) against baseline E_0:

$$1 + \Delta = \frac{1 + \chi^{E_1}}{1 + \chi^{E_0}}, \qquad 1 + \Delta_k = \frac{1 + \chi_k^{E_1}}{1 + \chi_k^{E_0}}, \quad k \in \{D, V, T\}, \qquad 1 + \Delta = (1 + \Delta_D)(1 + \Delta_V)(1 + \Delta_T).$$

Measuring Welfare

back

- **The object:** the benevolent government's value $V(y, b_D, b_F)$, CRRA with $\gamma = 2$.
- **A specification $\mathcal{E}$:** fixes $(\lambda_D, \lambda_F, \nu_D, \nu_F)$.
- **What we report:** the consumption change leaving the household indifferent,

$$(1+\Delta)^{1-\gamma}\,\bar{V}^{\mathcal{E}_0} = \bar{V}^{\mathcal{E}_1} \quad \Longleftrightarrow \quad \Delta_{\mathcal{E}_0 \to \mathcal{E}_1} = \left[\frac{\bar{V}^{\mathcal{E}_1}}{\bar{V}^{\mathcal{E}_0}}\right]^{\frac{1}{1-\gamma}} - 1, \qquad \bar{V} \equiv \textstyle\sum_y \pi(y)\, V(y, b_D, b_F).$$

- **Read it as:** the permanent raise making the baseline as good as the new economy.
- **Where it is evaluated:** zero debt.
- **For scale:** removing aggregate risk is worth **0.23** percent.

Welfare Decomposition: Policies Fixed

back

$$\hat{q}_i^D(y, b_D, b_F) = \frac{1-\theta}{1+r}\,\mathbb{E}_{y'|y}\Big[\hat{q}_i^D(y', b_D, b_F)\Big] + \theta\,\nu_i\,\hat{q}_i^{ND}(y, \nu_D b_D,\ \nu_F b_F)$$

- **Higher recovery:** a higher ν_i raises $\hat{q}_i^D$, and so $\hat{q}_i^{ND}$.
- **Higher debt at re-entry:** it re-enters owing $\nu_i b_i$.
- **Opposite signs, so we run all four combinations:**

run	prices $\hat{q}$	write-down $\nu_i b_i$	policies	what it isolates
D	baseline	baseline	fixed	Baseline
B	new	baseline	fixed	**Prices** $\Big\}\approx A$
C	baseline	new	fixed	**Re-entry** $\Big\}\approx A$
A	new	new	fixed	Prices + Re-entry
Full	new	new	**re-optimised**	Prices + Re-entry + **Behaviour**

All Four Experiments

back

- **Behaviour fixed at baseline policies**

	Δ spread (bp)		Δ welfare (%)		
	Domestic	Foreign	Pricing	Re-entry	Total
$\uparrow \nu_D$	−25.69	+0.44	+0.021	−0.013	+0.008
$\downarrow \nu_F$	−0.22	+26.32	−0.013	+0.007	−0.006
$\downarrow \nu_D$	+26.29	−0.28	−0.021	+0.013	−0.008
$\uparrow \nu_F$	+0.38	−25.56	+0.013	−0.007	+0.006

- **New recoveries, new policies**

	Δ spread (bp)		Debt / GDP	Foreign share	Default prob.	Δ welfare
	Domestic	Foreign	(%)	(%)	(%)	(%)
Baseline			20.7	30.5	3.3	
$\uparrow \nu_D$	−46.28	−22.35	21.1	0.3	3.0	+0.082
$\downarrow \nu_F$	−41.90	−17.91	18.8	0.3	2.8	−0.001
$\downarrow \nu_D$	+163.38	+112.59	24.6	83.0	4.6	−0.064
$\uparrow \nu_F$	+198.91	+131.25	28.9	89.3	5.4	+0.007

Same Maturities, Different Recoveries

back

Foreign-law share (%)

Recoveries	**Maturities**: both bonds (years)				
ν_D / ν_F	1.00	2.00	3.75	5.50	8.71
0.200 / 0.400	100.0	100.0	100.0	100.0	100.0
0.400 / 0.420	100.0	100.0	100.0	100.0	100.0
0.400 / 0.471	100.0	100.0	100.0	100.0	100.0
0.400 / 0.550	100.0	100.0	100.0	100.0	100.0
0.400 / 0.700	100.0	100.0	100.0	100.0	100.0
0.471 / 0.400	0.0	0.0	0.0	0.0	0.0

The last row reverses which bond has the higher recovery: the sovereign holds only the higher-recovery bond, even for a gap of 0.02.

Same Maturities, Different Recoveries

back

Foreign-law share (%), both bonds 3.75 years

Recoveries	ν_F				
ν_D	0.300	0.375	0.450	0.525	0.600
0.300	39.7	100.0	100.0	100.0	100.0
0.375	0.0	39.7	100.0	100.0	100.0
0.450	0.0	0.0	45.6	100.0	100.0
0.525	0.0	0.0	0.0	43.7	100.0
0.600	0.0	0.0	0.0	0.0	39.1

Shaded: equal recoveries. Off the diagonal, the sovereign holds only the higher-recovery bond.

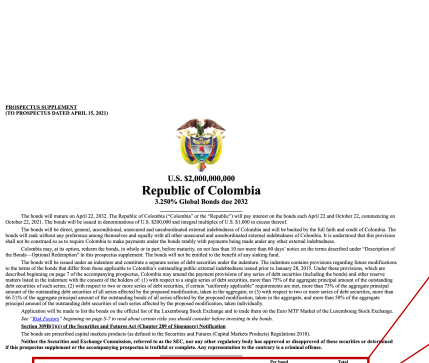
Same Recoveries, Different Maturities

back

Recoveries	**Maturities**: domestic / foreign duration (years)		
$\nu_D = \nu_F$	1 / 8.71	2 / 8.71	3.75 / 8.71 (crown pair)
0.00	47.8%	2.0%	0.3%
0.40	58.8%	2.0%	0.3%
0.4715	60.4%	2.0%	0.1%
0.60	65.3%	2.0%	0.1%
0.75	67.8%	3.7%	0.1%

Issuance costs: a foreign-law prospectus

back



PROSPECTUS SUPPLEMENT
(TO PROSPECTUS DATED APRIL 15, 2021)

U.S. $2,000,000,000

Republic of Colombia

3.250% Global Bonds due 2032

The bonds will mature on April 22, 2032. The Republic of Colombia ("Colombia" or the "Republic") will pay interest on the bonds each April 22 and October 22, commencing on October 22, 2021. The bonds will be issued in denominations of U.S. $200,000 and integral multiples of U.S. $1,000 in excess thereof.

The bonds will be direct, general, unconditional, unsecured and unsubordinated external indebtedness of Colombia and will be backed by the full faith and credit of Colombia. The bonds will rank equally in right of payment with all other direct, general, unconditional, unsecured and unsubordinated external indebtedness. It is understood that this provision shall not be construed so as to require Colombia to make payments under the bonds ratably with payments being made under any other external indebtedness.

Colombia may, at its option, redeem the bonds, in whole or in part, before maturity, on not less than 10 nor more than 60 days' notice on the terms described under "Description of the Bonds—Optional Redemption" in this prospectus supplement. The bonds will not be entitled to the benefit of any sinking fund.

The bonds will be issued under an indenture and constitute a separate series of debt securities under the indenture. The indenture contains provisions regarding future modifications to the terms of the bonds that differ from those applicable to Colombia's outstanding public external indebtedness issued prior to January 28, 2015. Under those provisions, which are described beginning on page 7 of the accompanying prospectus, Colombia may amend the payment provisions of any series of debt securities (including the bonds) and other reserve matters listed in the indenture with the consent of the holders of: (1) with respect to a single series of debt securities, more than 75% of the aggregate principal amount of the outstanding debt securities of such series; (2) with respect to two or more series of debt securities, if certain "uniformly applicable" requirements are met, more than 75% of the aggregate principal amount of the outstanding debt securities of all series affected by the proposed modification, taken in the aggregate; or (3) with respect to two or more series of debt securities, more than 66⅔% of the aggregate principal amount of the outstanding bonds of all series affected by the proposed modification, taken in the aggregate, and more than 50% of the aggregate principal amount of the outstanding bonds of each series affected by the proposed modification, taken individually.

Application will be made to list the bonds on the official list of the Luxembourg Stock Exchange and to trade them on the Euro MTF Market of the Luxembourg Stock Exchange.

See "Risk Factors" beginning on page S-7 to read about certain risks you should consider before investing in the bonds.

Section 309(B)(1)(c) of the Securities and Futures Act (Chapter 289 of Singapore) Notification

The bonds are prescribed capital markets products (as defined in the Securities and Futures (Capital Markets Products) Regulations 2018).

Neither the Securities and Exchange Commission, referred to as the SEC, nor any other regulatory body has approved or disapproved of these securities or determined if this prospectus supplement or the accompanying prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

	Per bond	Total
Public offering price(1)	99.032%	U.S. $1,980,640,000
Underwriting discount	0.200%	U.S. $ 4,000,000
Proceeds, before expenses, to Colombia	98.832%	U.S. $1,976,640,000

(1) Purchasers will also be required to pay accrued interest, if any, from April 22, 2021, if settlement occurs after that date.

Source: Republic of Colombia, prospectus supplement (424B5) for the U.S.$2,000,000,000 3.250% Global Bonds due 2032, April 2021. SEC EDGAR accession 0001193125-21-124194, cover page. Highlight added.

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Proceeds, before expenses, to Colombia	98.832%	U.S. $1,976,640,000

Issuance Costs Differ by Governing Law

back

- **Issuance costs likely differ substantially.**
  - Domestic-law bonds are sold through competitive auctions (central bank).
  - Foreign-law bonds are placed through syndication ($\approx$20 bp of face).
    - Requires more time and coordination.
    - Requires legal advisors and bookrunners.

- **One-period bonds:** foreign-law issuance stops at $\approx$**75 bp**, not 20 bp.